

SG/Send Data Room Architecture

Version 0.5.10 | Architecture Brief



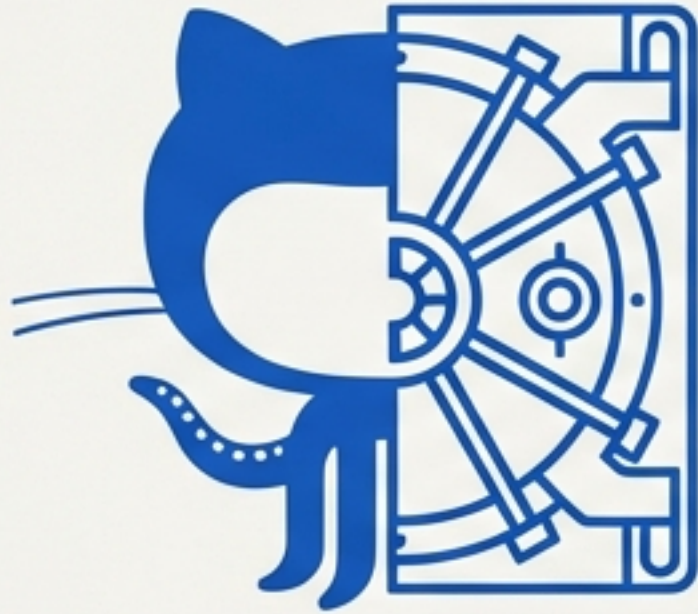
Date: 21 Feb 2026

From: Human (Project Lead)

To: Architect, Developer, DevOps

Ref: Internal Architecture Brief

Infrastructure that exists only when needed. Data that is auditable forever.



Storage (The Static State)

GitHub-backed Persistence.
Configuration and structure are
version-controlled and permanent.



Compute (The Active State)

Ephemeral / Scale-to-Zero.
Servers are disposable entities
that spin up on demand.



Security (The Boundary)

Zero-Trust & Encrypted.
'Sealed Box' architecture with
no egress and strict isolation.

A paradigm shift from always-on servers to on-demand materialisation.

Two Classes of Truth: Persistent vs. Ephemeral

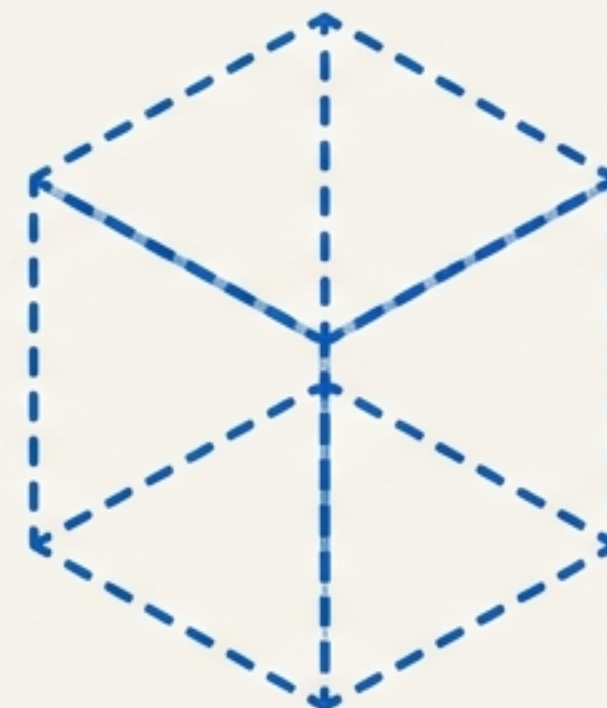
Persistent / GitHub



- **What:** Configuration, branding, persona definitions.
- **Durability:** Permanent, auditable, recover.
- **Source of Truth:** GitHub IS the truth.
- **Encryption:** Cleartext in repo, keys in Secrets.

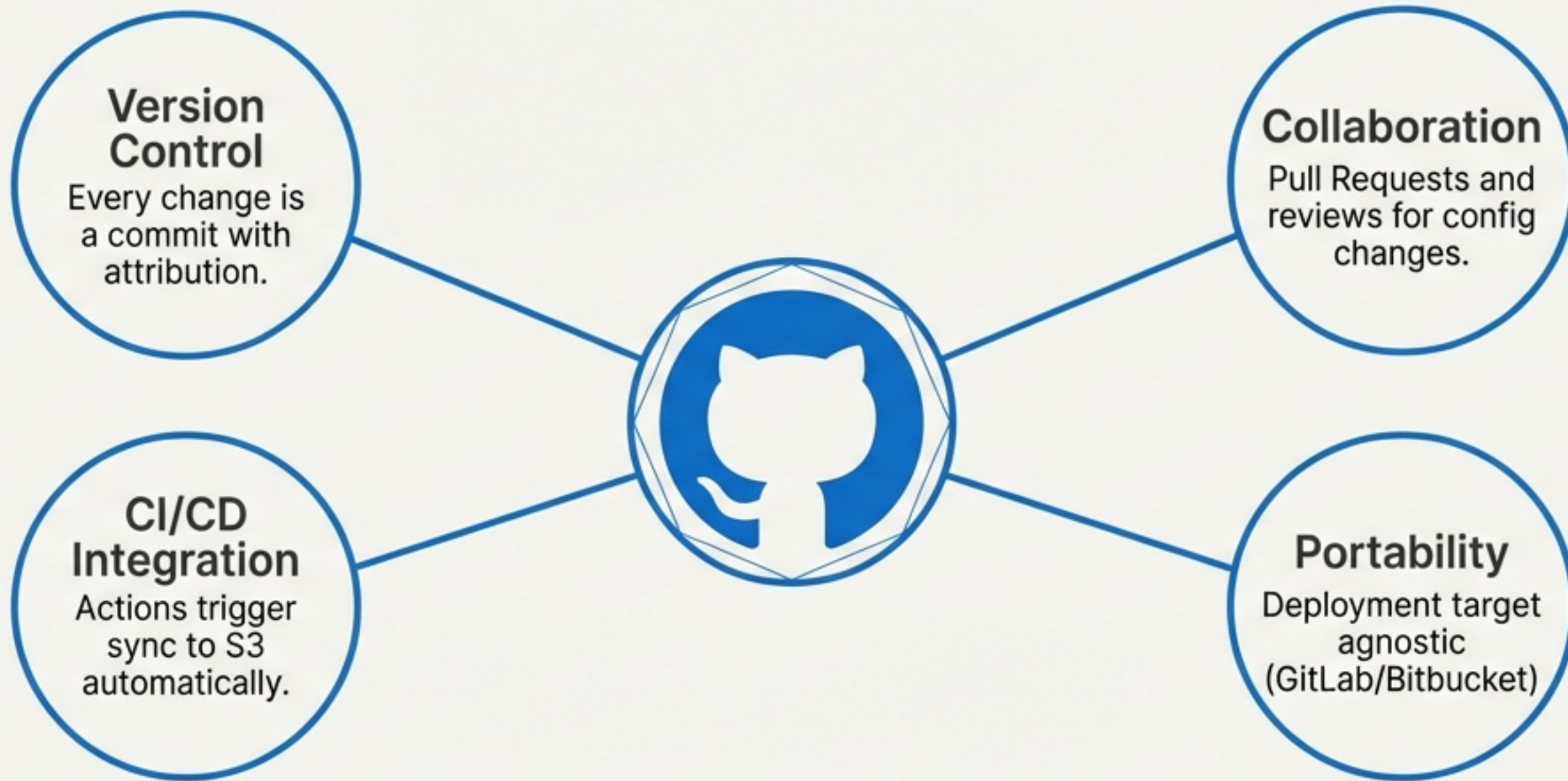
GitHub holds the history.
S3 holds the secrets.

Ephemeral / S3



- **What:** Encrypted file transfers, messages.
- **Durability:** Disposable with TTL (hour/day).
- **Source of Truth:** S3 is merely a cache.
- **Encryption:** Always encrypted (PKI).

GitHub is the Store of Record



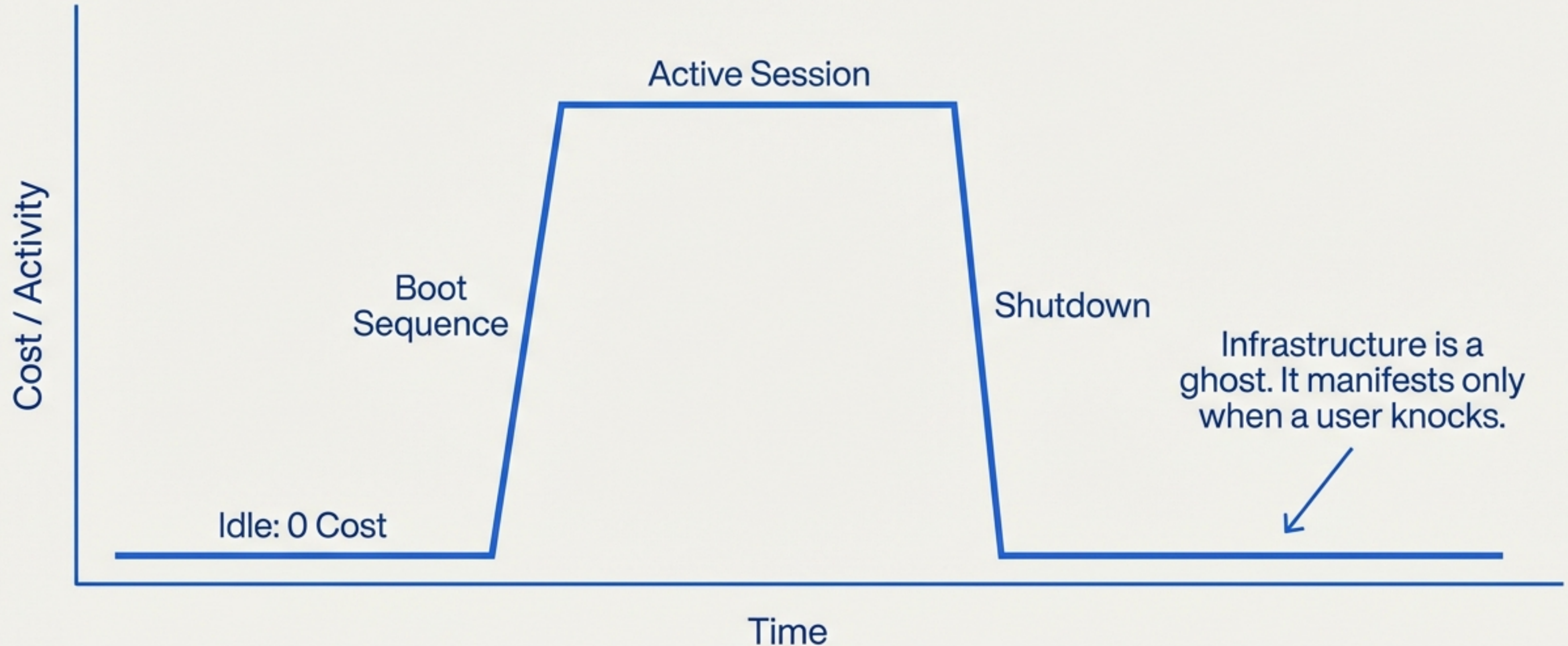
S3 buckets can be deleted or misconfigured. A git repository provides a complex, reversible history book that simple storage cannot match. Moving from email to GitHub as a data store is a massive upgrade in auditability.

The Sync Pipeline: Secure Transition to Storage

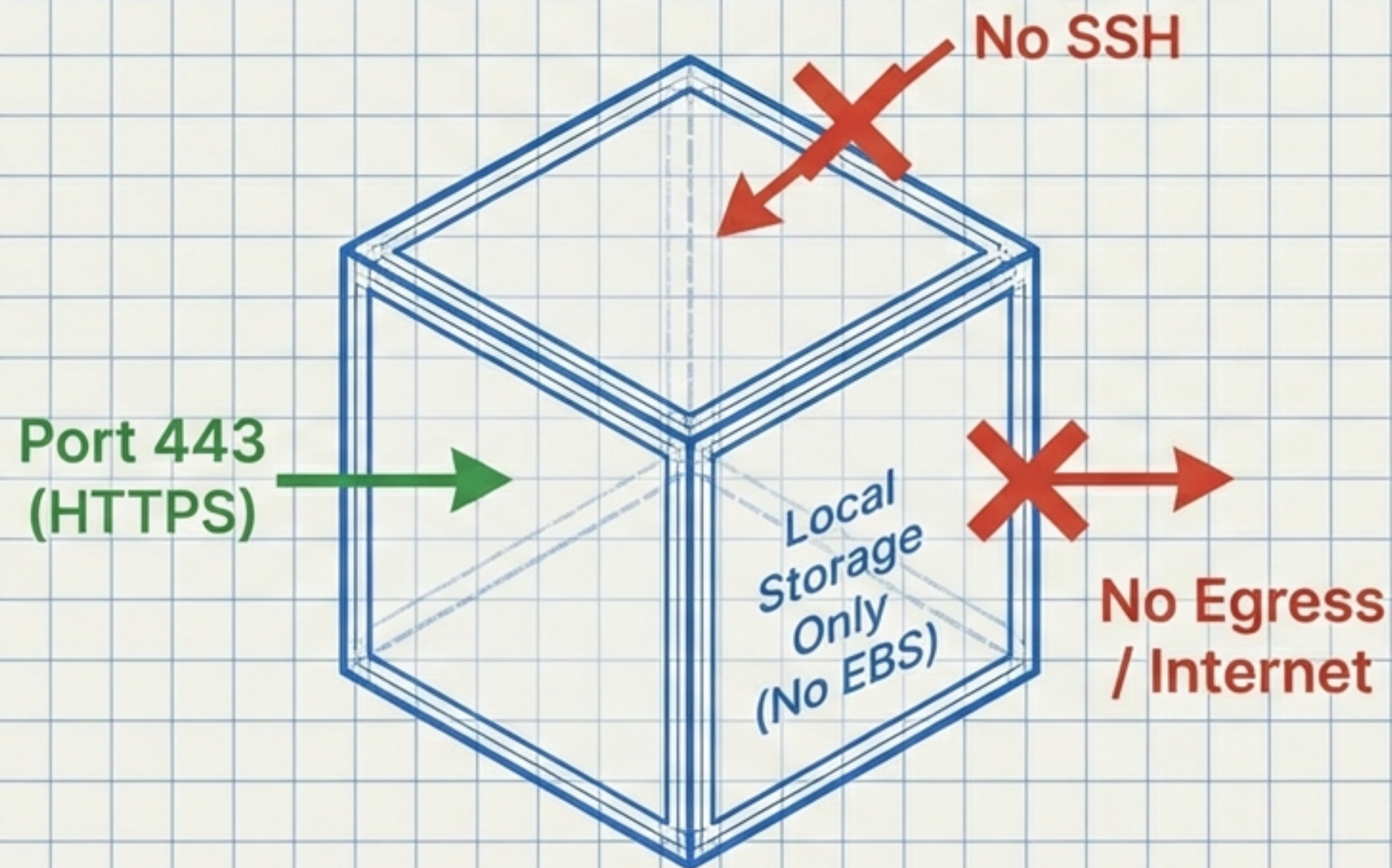


CRITICAL: Private keys are **NEVER** in the repo.
They exist only in GitHub Secrets or user browsers.

Ephemeral Compute and Scale-to-Zero Economics

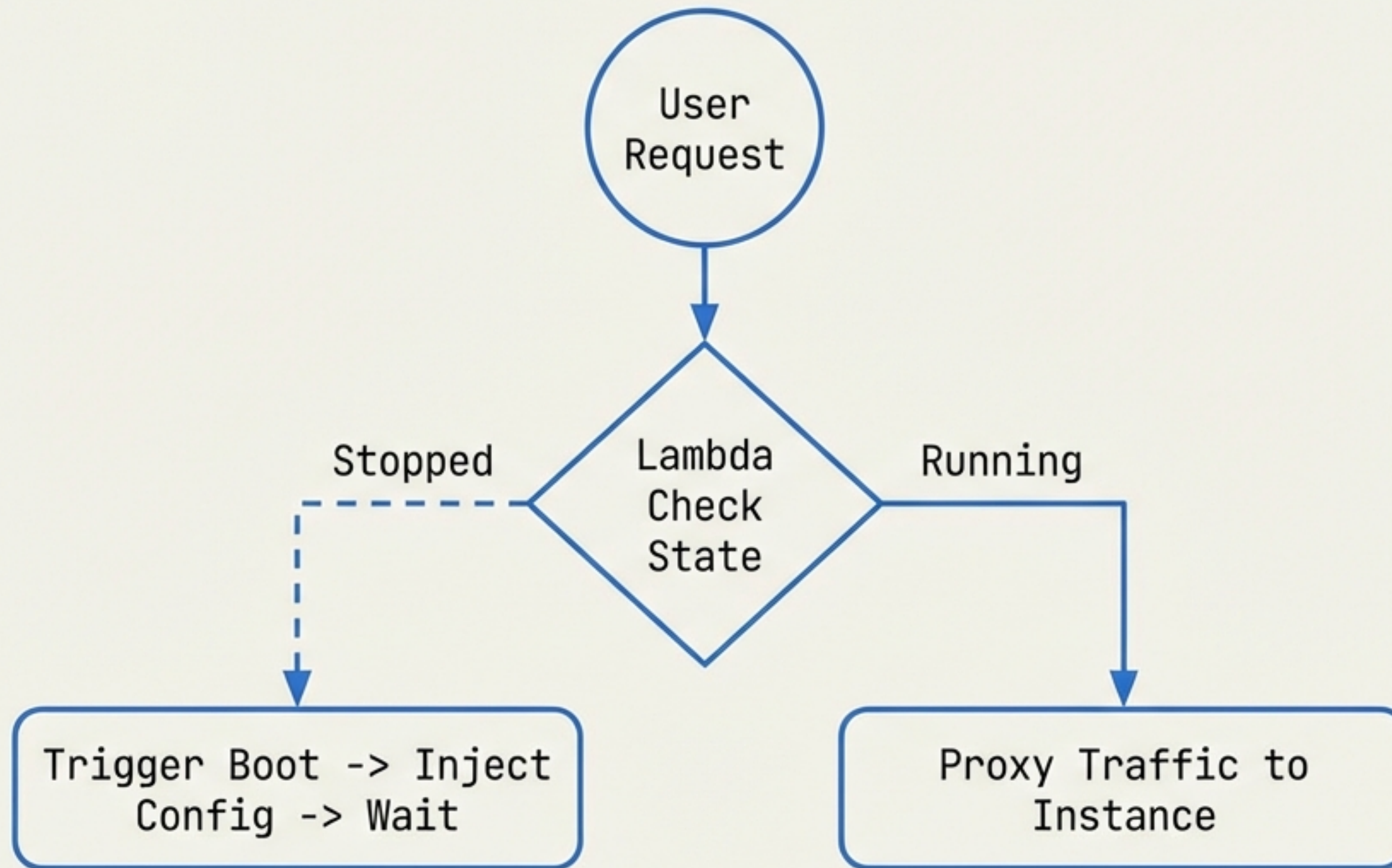


The “Sealed Box” Instance Architecture



- **Data Loading:** PUSH only via Admin API.
- **Admin Access:** One-time token on boot.

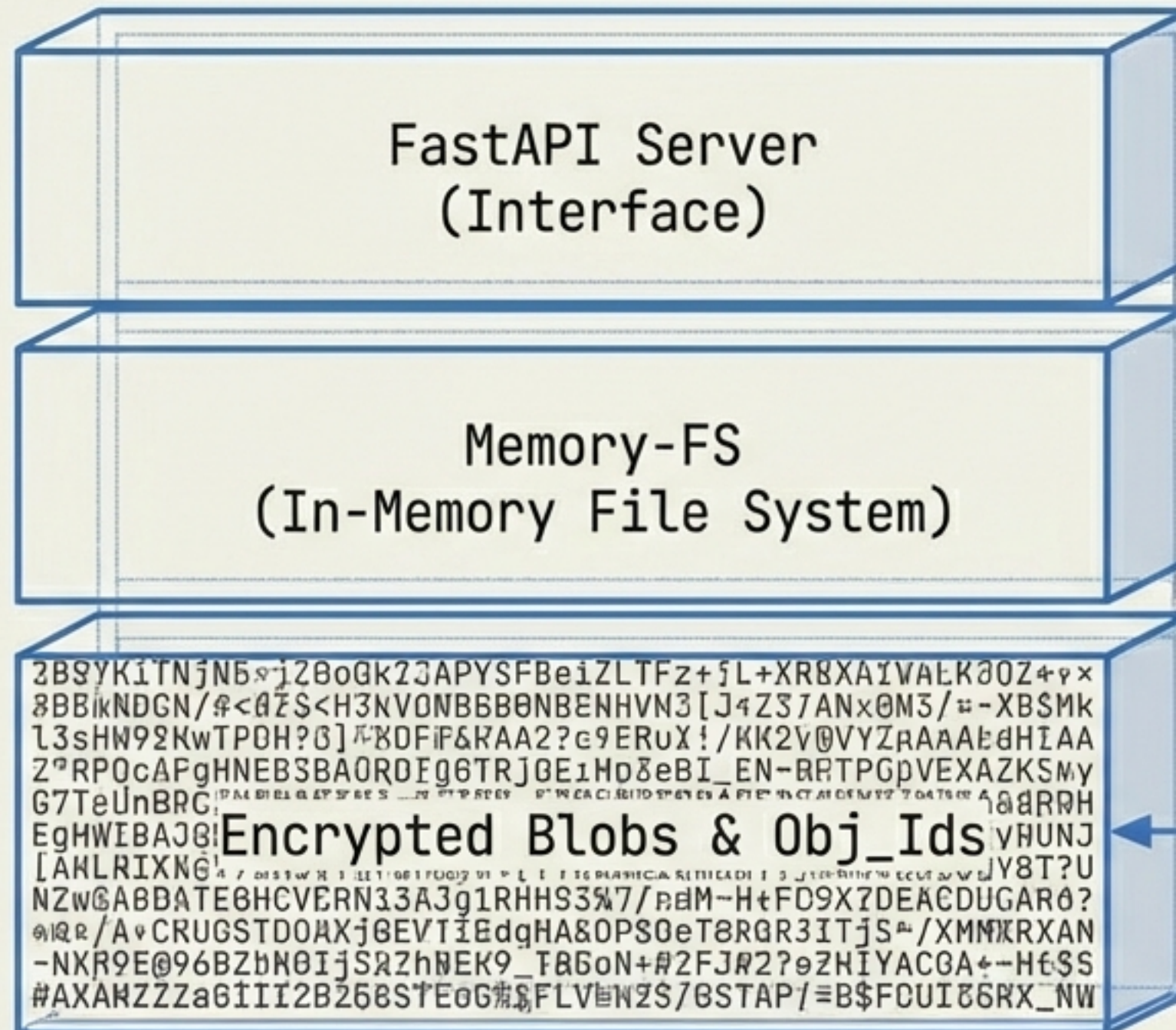
The Orchestration Machine



Lambda handles the logic; EC2 provides the stateful compute.

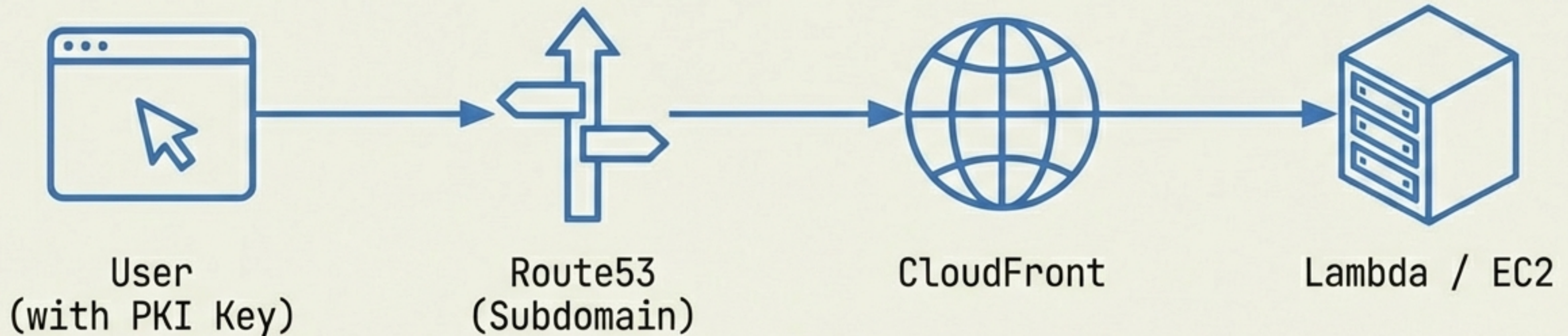
We use EC2 over Lambda for the runtime to guarantee in-memory file system state.

Runtime Internals: Memory-FS and Encrypted Blobs



The server operates on encrypted blobs. It sees no plaintext strings and holds no metadata in cleartext.

Connectivity, DNS, and Routing



DevOps Note: DNS entries and CloudFront distributions are provisioned automatically via the Fleet API upon room creation.

The Fleet Management API

POST `/api/fleet/rooms` :
Create new room (Provisions DNS, Links Repo)

POST `/api/fleet/rooms/{id}/start` :
Boot instance

POST `/api/fleet/rooms/{id}/stop` :
Shutdown (syncs ephemeral data first)

DELETE `/api/fleet/rooms/{id}` :
Decommission (terminate & cleanup)

POST `/api/fleet/rooms/{id}/sync` : Trigger GitHub -> S3 sync
Trigger GitHub -> S3 sync

“All operations are API-first. The Admin UI is just a convenience layer.”

Future Roadmap: Cryptographic Attribution



Concept: Write-back Capability.

Mechanism: Commits are signed with the user's PKI key.

Result: Every change to the configuration is traceable to a specific, authenticated identity within the git history.

Cost Model and Efficiency

Variable Costs (Usage-Based)	Fixed Costs (Base)	Negligible
• EC2 Compute (Active time only) • S3 Storage (Per GB) • CloudFront (Transfer)	• DNS (Route53 ~ \$0.50/month per room)	• Lambda (Boot/Shutdown logic)

Cost per data room per month is driven by active usage, not existence. Idle rooms cost pennies.

Architectural Principles

■ 1. GitHub is Truth

Configuration is versioned, auditable, and persistent.

■ 2. Compute is Ephemeral

Infrastructure is a temporary state container.

■ 3. Network is Zero-Trust

Inbound 443 only. No egress. No SSH.

■ 4. Cost is Usage-Based

Zero idle cost beyond storage and DNS.

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